

# MongoDB Placement Guide

## 1. Introduction to MongoDB

NoSQL, document-based database, schema-less, scalable.

Advantages: Flexible, High performance, Easy to scale.

SQL vs NoSQL comparison.

Example:

```
db.users.insertOne({name: 'Pankaj', age: 22, course: 'Node.js'});
```

## 2. MongoDB Architecture

Hierarchy: Database > Collection > Document > Field.

Example document:

```
{
  '_id': 1,
  'name': 'Ravi',
  'course': 'MERN Stack',
  'marks': 92
}
```

## 3. CRUD Operations

Create: insertOne, insertMany.

Read: find(), projection.

Update: updateOne(), updateMany().

Delete: deleteOne(), deleteMany().

Examples included.

## 4. Query Operators

\$eq, \$gt, \$lt, \$in, \$and, \$or.

Example:

```
db.students.find({$and:[{marks:{$gt:80}}, {city:'Delhi'}]})
```

## 5. Indexing

Single, Compound, Unique, Text.

Example:

```
db.students.createIndex({name:1})
```

## 6. Aggregation Framework

Stages: \$match, \$group, \$project, \$sort, \$limit.

Example:

```
db.students.aggregate([{$group:{_id:'$course', avgMarks:{$avg:'$marks'}}}])
```

## **7. Relationships**

Embedded documents vs Referenced documents.  
Examples provided.

## **8. MongoDB with Node.js**

Connection with MongoClient.  
CRUD examples using async/await.

## **9. Advanced Topics**

Replication & Replica Sets, Sharding, Transactions, GridFS, Security and Authentication.  
Examples included.

## **10. Solved Interview Questions**

20+ common questions with solutions and query examples.

## **11. Data Modeling Tips**

Embedded vs Referenced.  
Best practices and examples.

## **12. Last-Minute Revision Tips**

Key commands and topics for interview readiness.